

Aditeya Baral

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EDUCATION

New York University, Courant Institute of Mathematical Sciences

Masters in Computer Science; GPA - 3.87/4.00

Concentration: Artificial Intelligence

New York City, USA

Sep 2024 – May 2026

PES University

Bachelor of Technology in Computer Science & Engineering; GPA - 8.71/10.00

Specialization: Machine Intelligence & Data Science

Bengaluru, India

Aug 2018 – May 2022

EXPERIENCE

Redis

San Francisco, USA

Applied Research Scientist Intern, Redis LangCache; Advisor: Srijith Rajamohan

June 2025 – Dec 2025

- Architected a *two-stage retrieval and re-ranking* pipeline for **Redis LangCache**, achieving a **12.5% PR-AUC** and **8% P-CHR AUC** improvement over baselines by integrating *cross-encoder re-rankers* for *full token-level interaction*.
- Curated and open-sourced **LangCache SentencePairs (v1-v3)**, a large-scale dataset family spanning **1M to 40M examples** from diverse linguistic sources, enabling robust fine-tuning of semantic retrieval and re-ranking models.
- Open-sourced **LangCache ReRanker v1** and **v2** model families comprising cross-encoder variants fine-tuned with ranking and classification objectives, enabling *application-specific score calibration* for diverse semantic caching use cases.
- Assisted in the fine-tuning and deployment of **LangCache Embed v3**, a generalist model for semantic retrieval, achieving **13.5% PR-AUC** improvement over v2 and outperforming larger general-purpose models even without re-ranking.
- Developed a *comprehensive evaluation framework* integrated with **RedisVL** for LangCache customers, enabling systematic analysis of achievable P-CHR tradeoffs, valid cache-hit rates, and operational thresholds before onboarding.
- Quantified *retriever bottlenecks* and *aggressive vs. conservative re-ranking effectiveness* by analyzing recall ceilings and re-ranking movement to *optimize operational trade-offs* and *improve cache-hit quality*.
- Supported *downstream integration* and development of **LMCache** by *building prototypes* and *conducting performance studies* with **Redis** as an *in-memory KV store*, demonstrating latency and throughput gains.

Cisco Systems

Bengaluru, India

Applied AI Engineer, Webex Media Quality Analytics

July 2022 – July 2024

- Instruction fine-tuned LLMs like **Mistral** and **Llama-2** on-prem to enable *secure* and *cost-effective* AI solutions such as *translation* and *RAG* for engineers and customers, *cutting third-party dependency costs by 30%*.
- Led the initiative to build a novel *pre-training algorithm* for conversational data using **PyTorch** and **HuggingFace**, achieving a **40% performance gain** over standard approaches at benchmark fine-tuning tasks.
- Developed the **Webex Contextual Search** engine and *improved searching, ranking, recommendations* and *topic modelling* by **75%** with **<10%** increased overhead latency.
- Integrated **OpenAI** APIs and on-prem LLMs with the **Webex AI Assistant** for **15M+** worldwide users to add *auto-replies*, *summarisation*, *querying* and *action-item extraction* to message threads and meeting transcripts.

Big Data Engineering Intern, Webex VideoMesh Analytics

Jan 2022 – June 2022

- Migrated the **Meetings Analytics Engine** from Java and Spark to **Scala** and **Flink** to scale up to **1M+ reports/min** and significantly *improve real-time report generation* by over **40%**.
- Built **VideoMesh Developer APIs** using **Java** and globally rolled them out for **30,000+ enterprises** with **customer-facing applications**.

Intel Corporation

Bengaluru, India

Applied Research Scientist Intern, Intel VSG; Advisors: Anay Majee, Anbumani Subramanian

Aug 2021 – Dec 2021

- Explored **Few-Shot Learning Object Detection (FSOD)** techniques to reduce *catastrophic forgetting* in constrained and heterogeneous driving environments.
- Investigated and designed novel *representation learning* and *attention mechanisms* to learn *inter/intra-object relationships* using **PyTorch**.
- Outperformed existing approaches at the time on base and novel classes by **0.2 mAP** and **3 mAP** on the *Few-Shot India Driving Dataset*, a benchmark for FSOD.

SKILLS

Languages: Python, Scala, Java, C/C++, R, Groovy, SQL, LaTeX

ML/Stats Libraries: PyTorch, Tensorflow, HuggingFace, NLTK, pandas, NumPy, scikit-learn, seaborn, matplotlib, plotly

AI/ML Techniques: Representation Learning, Mechanistic Interpretability, Transfer Learning, Language Models, RAG

Big Data/Cloud: Hadoop, Kafka, Zookeeper, Spark, Flink, Iceberg, Pinot, Redis, ELK

Frameworks/Tools: Git, GitHub, Jenkins, Docker, Kubernetes, Flask, Grafana, PSQ, MongoDB, AWS, Linux

- [1] **When ‘+’ Means ‘-’? Probing Arithmetic Circuits Under Symbol Redefinition**
Authors: Aditeya Baral, Allen George Ajith, Shauli Ravfogel
- [2] **Can LLMs *understand* Math? Exploring the Pitfalls in Mathematical Reasoning**
Authors: Tiasa Singha Roy, Aditeya Baral*, Ayush Rajesh Jhaveri, Yusuf Baig*
- [3] **CMLFormer – A Dual Decoder Transformer with Switching Point Learning for Code-Mixed Language Modeling**
Authors: Aditeya Baral, Allen George Ajith, Roshan Nayak, Mrityunjay Abhijeet Bhanja
- [4] **Patch and Control – Steering Behavior of Large Vision-Language Models via Latent Activations**
Authors: Aditeya Baral, Rijul Dahiya, Dilip Venkatesh

PAPERS AND PUBLICATIONS

- [1] **Training for Compositional Sensitivity Reduces Dense Retrieval Generalization**
ICLR 2026, Sci4DL
Authors: Radoslav Ralev, Aditeya Baral, Iliya Zhechev, Jen Agarwal, Srijith Rajamohan
- [2] **ChatBERT – Multi-task approach to Pre-Training for Structured Conversations**
Webex AI 2023
Authors: Aditeya Baral (Work done as part of Cisco Webex AI Research)
- [3] **CalBERT – Code-mixed Adaptive Language Representations using BERT**
AAAI 2022, MAKE
Authors: Aditeya Baral, Aronya Baksy, Ansh Sarkar, Deeksha D, Ashwini M Joshi
- [4] **Information Maximization to Overcome Catastrophic Forgetting in Few-Shot Object Detection**
Intel VSG Research 2021
Authors: Aditeya Baral, Anay Majee, Anbumani Subramanian
- [5] **MAPLE – MAsking words to generate blackout Poetry using Seq2Seq LEarning**
ACL 2021, ICNLSP
Authors: Aditeya Baral, Himanshu Jain, Deeksha D, Mamatha H R
- [6] **Analysis of Kepler Objects of Interest using ML for Exoplanet Identification**
IEEE CONIT 2021
Authors: Ameya Rajendra Bhamare, Aditeya Baral, Saarthak Agarwal